

Chapter 14.

14.1 Proof: $X_k = AX_{k-1} = A(AX_{k-2}) = A^2X_{k-2}$
 $= \dots = A^k X_0$

14.2. Trivial.

14.3. $\lambda_1 = 4, \lambda_2 = 1$

14.4 1st loop: $A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$ $X_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ let $X = X_0$

$$Z = \frac{X}{\|X\|_2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, X = AZ = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \alpha = Z^T X = 2$$

2nd loop:

$$Z = \frac{X}{\|X\|_2} = \begin{bmatrix} 0.821 \\ 0.5547 \end{bmatrix}, X = AZ = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 0.821 \\ 0.5547 \end{bmatrix} = \begin{bmatrix} 3.3282 \\ 3.6056 \end{bmatrix}, \alpha = Z^T X = 4.7692$$

3rd loop:

$$Z = \frac{X}{\|X\|_2} = \begin{bmatrix} 0.6783 \\ 0.7348 \end{bmatrix}, X = AZ = \begin{bmatrix} 3.5610 \\ 3.5044 \end{bmatrix}, \alpha = Z^T X = 4.9904.$$

It converges to the dominant eigenvalue $\lambda = 5$ and the relative error after 3rd loop is: $\frac{5 - 4.9904}{5} = 0.0019$

14.5. $\lambda_1(A) = 5, V_1 = C_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda_2(A) = -1, V_2 = C_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

If we take $X_0 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ which is an eigenvector of A corresponding to the eigenvalue $\lambda_2 = -1$, then the Power method fails to converge.

~~Since $AZ = A \frac{X}{\|X\|_2} = 0$ always. Since $\alpha = 0$ at all iterations.~~

14.6. $A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$ $X_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ let $X = X_0$.

1st loop:

$$Z = \frac{X}{\|X\|_2} = \begin{bmatrix} 0.5774 \\ 0.5774 \\ 0.5774 \end{bmatrix}, X = AZ = \begin{bmatrix} 2.3094 \\ 2.3094 \\ 1.1547 \end{bmatrix}, \alpha = Z^T X = 3.3333$$

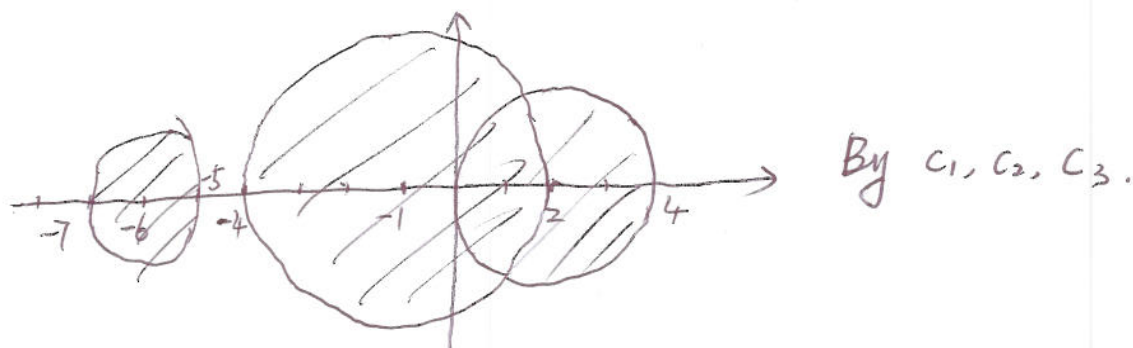
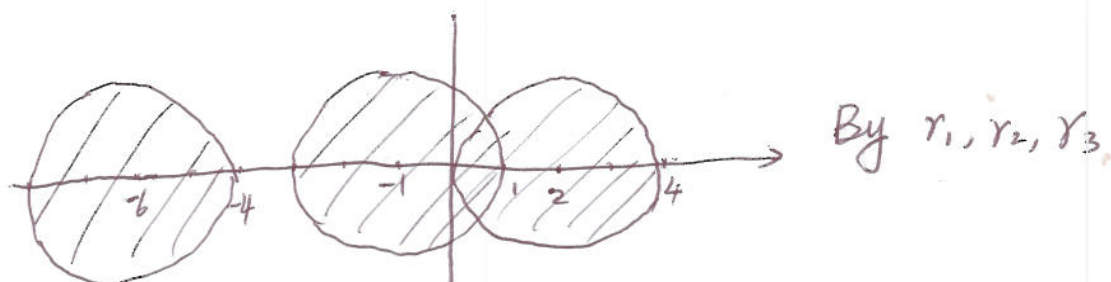
2nd loop:

$$Z = \frac{x}{\|x\|_2} = \begin{bmatrix} 0.6667 \\ 0.6667 \\ 0.3333 \end{bmatrix}, \quad x = AZ = \begin{bmatrix} 1.3333 \\ 3.0 \\ 1.6667 \end{bmatrix}, \quad \alpha = Z^T x = 3.4444.$$

3rd loop:

$$Z = \frac{x}{\|x\|_2} = \begin{bmatrix} 0.3621 \\ 0.8148 \\ 0.4527 \end{bmatrix}, \quad x = AZ = \begin{bmatrix} 1.3580 \\ 2.2636 \\ 1.0864 \end{bmatrix}, \quad \alpha = Z^T x = 2.8279.$$

14.7. $a_{11}=2, \quad a_{22}=-1, \quad a_{33}=-6$
 $r_1=2, \quad r_2=2, \quad r_3=2.$
 $c_1=2, \quad c_2=3, \quad c_3=1$



There is a dominant eigenvalue $-7 \leq \lambda \leq -5$.

14.8. There is a real eigenvalue λ_1 such that $4 \leq \lambda_1 \leq 6$
 a real eigenvalue λ_2 such that $-5 \leq \lambda_2 \leq -1$

There are two eigenvalues λ_3, λ_4 in the disc $B(1, 1)$.

14.9. Using the inverse power method. Since $|\lambda_1| \leq |\lambda_j|$ for $j=2, 3$.

14.10. Using the shifted Inverse Power method by taking $s=1$ since the eigenvalue λ_1 inside that disc satisfying

$$|\lambda_1 + 1| < |\lambda - 1| \text{ for any other eigenvalue } \lambda.$$

14.11. Using the shifted Inverse Power method by taking $s=6$.

14.12. $A = \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}$ $X = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

1) Using Power method to compute the dominant eigenvalue.

1st loop:

$$Z = [1 \ 0 \ 0 \ 0 \ 0]^T, \quad X = AZ = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \alpha = Z^T X = 2.$$

2nd loop

$$Z = \frac{X}{\|X\|_2} = \begin{bmatrix} 0.8944 \\ -0.4472 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad X = AZ = \begin{bmatrix} 2.2361 \\ -1.7889 \\ 0.4472 \\ 0 \\ 0 \end{bmatrix}, \quad \alpha = 2.800.$$

$$\frac{|2.8 - 2|}{2.8} = 0.2857 > 0.1, \text{ Continue.}$$

3rd loop:

$$Z = \frac{X}{\|X\|_2} = \begin{bmatrix} 0.7715 \\ -0.6172 \\ 0.1543 \\ 0 \\ 0 \end{bmatrix}, \quad X = AZ = \begin{bmatrix} 2.1602 \\ -2.1602 \\ 0.9258 \\ -0.1543 \\ 0 \end{bmatrix}, \quad \alpha = 3.1429.$$

$$\frac{|3.1429 - 2.8|}{3.1429} = 0.1091 > 0.1, \text{ Continue.}$$

4th loop:

$$Z = \frac{X}{\|X\|_2} = \begin{bmatrix} 0.6759 \\ -0.6759 \\ 0.2897 \\ -0.0483 \\ 0 \end{bmatrix}, \quad X = AZ = \begin{bmatrix} 2.0278 \\ -2.3175 \\ 1.3036 \\ -0.3862 \\ 0.0483 \end{bmatrix}, \quad \alpha = 3.3333.$$

$$\frac{|3.3333 - 3.1429|}{3.3333} = 0.0571 < 0.1, \text{ stop.}$$

2) Using the inverse power method to find the eigenvalue closer to 0 than the others. (skip this step in your homework).